SOCIETY OF MARINE ENGIEERING STUDENT

In Collaboration With UNESCO JKUAT

LESSON HANDOUT

LaTeX Capabilities Showcase

A Comprehensive Reference for Students

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This document is a comprehensive showcase of LaTeX's typesetting capabilities, intended for university students beginning to learn LaTeX

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1 Text Formatting

1.1 Basic Text Styles

LaTeX provides many ways to style text. The most common are:

- **Bold**: `\textbf{text}`
- *Italic*: `\textit{text}`
- `Monospace`: `\texttt{text}`
- Underline: `\underline{text}`
- SMALL CAPS: `\textsc{text}`
- *Emphasis* (context-aware): `\emph{text}`
- Larger or Smaller text via size declarations
- Coloured text via `\textcolor{colour}{text}`

1.2 Font Size Hierarchy

LaTeX has a built-in size hierarchy (from smallest to largest):

tiny scriptsize footnotesize small normalsize large Large LARGE huge Huge

1.3 Text Alignment

This paragraph is **left-aligned** (flushleft environment). LaTeX normally justifies text to both margins, which is the typographic standard for books and academic papers.

This paragraph is **centred** (center environment).

This paragraph is **right-aligned** (flushright environment).

1.4 Quotations

For a short inline quotation, use `\enquote`: *"Typography is the art and technique of arranging type."*

For a longer block quotation, use the `quote` environment:

"Not everything that can be counted counts, and not everything that counts can be counted."
— William Bruce Cameron

1.5 Footnotes and Margin Notes

Footnotes[1] are easy in LaTeX. You can also add margin notes for editorial comments.

Margin note here.

1.6 Spacing

Horizontal space: A B (1 cm of explicit space between A and B).

Vertical space: use `\vspace{0.5cm}` to add vertical gaps.

A horizontal rule spanning the full width:

---

[1] This is a footnote. Use `\footnote{...}` to add one.

2 Lists

2.1 Unordered (Bullet) Lists

- First item
- Second item
  - Nested item A
  - Nested item B
    - Deep item i
    - Deep item ii
- Third item

2.2 Ordered (Numbered) Lists

1. First step
2. Second step
   a. Sub-step a
   b. Sub-step b
3. Third step

2.3 Description Lists

pdfLaTeX
: Compiles `.tex` to PDF; supports most packages; the most widely-used engine.

XeLaTeX
: Supports Unicode and OpenType fonts natively; great for multi-lingual documents.

LuaLaTeX
: Like XeLaTeX but also allows embedded Lua scripting; very flexible.

3 Mathematics

Mathematics is arguably LaTeX's greatest strength.

3.1 Inline and Display Mode

Inline maths is written between `$...$`: the famous formula is $e^{i\pi} + 1 = 0$ (Euler's identity).

Display (standalone) maths uses `\[...\]` or the `equation` environment:

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{1}$$

3.2 Fractions, Roots, and Sums

$$\sqrt[n]{a^n + b^n} \neq a + b \qquad \frac{\partial^2 u}{\partial t^2} = c^2\nabla^2 u \qquad \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}$$

3.3 Integrals and Limits

$$\int_{-\infty}^{+\infty} e^{-x^2}\,\mathrm{d}x = \sqrt{\pi} \qquad \lim_{x\to 0} \frac{\sin x}{x} = 1 \qquad \oint_{\mathcal{C}} \mathbf{F}\cdot\mathrm{d}\mathbf{r} = 0$$

3.4 Aligned Equations

The `align` environment aligns equations at the `&` symbol:

$$(a+b)^2 = a^2 + 2ab + b^2 \tag{2}$$
$$(a-b)^2 = a^2 - 2ab + b^2 \tag{3}$$
$$(a+b)(a-b) = a^2 - b^2 \tag{4}$$

We can reference numbered equations: Equation (1) is Euler's formula; Equation (2) expands a perfect square.

3.5 Matrices

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The determinant:

$$\det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

3.6 Theorems, Definitions, and Proofs

Definition 3.1. A function $f : \mathbb{R}^n \to \mathbb{R}$ is **convex** if for every $x, y \in \mathbb{R}^n$ and every $\lambda \in [0, 1]$:

$$f\big(\lambda x + (1-\lambda)y\big) \;\leq\; \lambda f(x) + (1-\lambda) f(y).$$

Theorem 3.2 (Cauchy–Schwarz Inequality)**.** *For all vectors* $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$*:*

$$\big|\langle \mathbf{u},\, \mathbf{v}\rangle\big|^2 \;\leq\; \langle \mathbf{u},\, \mathbf{u}\rangle \cdot \langle \mathbf{v},\, \mathbf{v}\rangle.$$

Proof. Consider the function $f(t) = \|\mathbf{u} + t\mathbf{v}\|^2 \geq 0$ for all $t \in \mathbb{R}$. Expanding and requiring the discriminant to be non-positive yields the result. □ □

Remark. Equality holds if and only if $\mathbf{u}$ and $\mathbf{v}$ are linearly dependent.

3.7 Special Symbols and Operators

The standard number sets defined by our custom macros: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{R} \subset \mathbb{C}$.

Greek letters appear frequently in mathematics: $\alpha, \beta, \gamma, \delta, \epsilon, \zeta, \eta, \theta, \lambda, \mu, \xi, \pi, \rho, \sigma, \tau, \phi, \chi, \psi, \omega$.

Common operators: $\forall$, $\exists$, $\in$, $\notin$, $\subset$, $\cup$, $\cap$, $\oplus$, $\otimes$, ∇, ∂, ∞, $\pm$, $\mp$, $\times$, $\div$, $\leq$, $\geq$, $\neq$, $\approx$, $\sim$, $\equiv$, $\propto$.

3.8 Piecewise Functions

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases}$$

4 Tables

4.1 Professional Table with Booktabs

The `booktabs` package removes vertical lines and uses horizontal rules to produce publication-quality tables (see Table 1).

Table 1. Comparison of common LaTeX engines

Engine	Font Support	Unicode	Speed
pdfLaTeX	Type-1, OTF (limited)	No	Fast
XeLaTeX	All OpenType	Yes	Moderate
LuaLaTeX	All OpenType	Yes	Slower

4.2 Coloured Table with Multirow Cells

Table 2 demonstrates row colours and merged cells.

Table 2. Grade distribution (sample data)

Grade Range	Letter	GPA Points
90 – 100	A+	4.0
80 – 89	A	3.7
70 – 79	B	3.0
60 – 69	C	2.0
50 – 59	D	1.0
<50	F	0.0

4.3 Auto-Width Columns with `tabularx`

Table 3. Common LaTeX environments and their purposes

Environment	Purpose
`equation`	Display a single numbered mathematical expression.
`align`	Align multi-line equations at a specific character.
`itemize`	Create unordered (bulleted) lists.
`enumerate`	Create ordered (numbered) lists.
`tabular`	Construct tables with rows and columns (tabularx for auto-width X-columns).
`figure`	Float a figure (image or TikZ drawing) on the page.
`lstlisting`	Display formatted source code.
`theorem`	State a mathematical theorem.

5 Figures and Graphics

5.1 Including External Images

Use `\includegraphics` to embed images. Supported formats include PNG, JPG, and PDF. The `width` option scales the image:

[Replace with `\includegraphics{your-image}`]

Figure 1. A placeholder showing where an external image would appear.

5.2 TikZ Diagrams

TikZ is a powerful in-document vector graphics library. Figure 2 shows a simple flowchart drawn entirely in LaTeX.

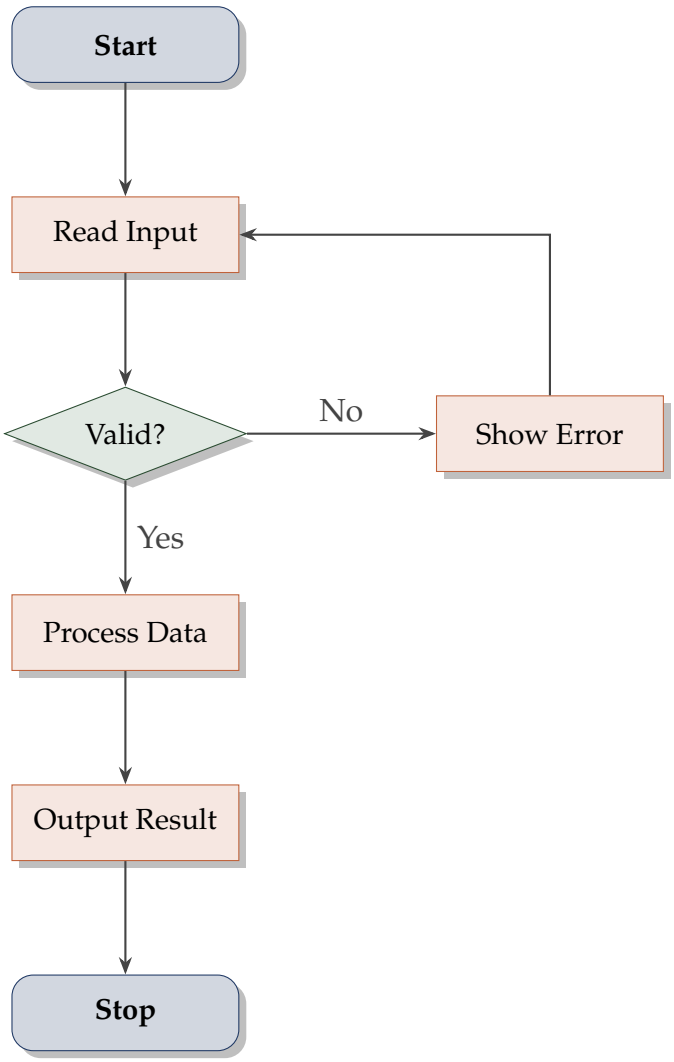

Figure 2. A flowchart drawn entirely with TikZ inside LaTeX.

5.3 Function Plot with TikZ

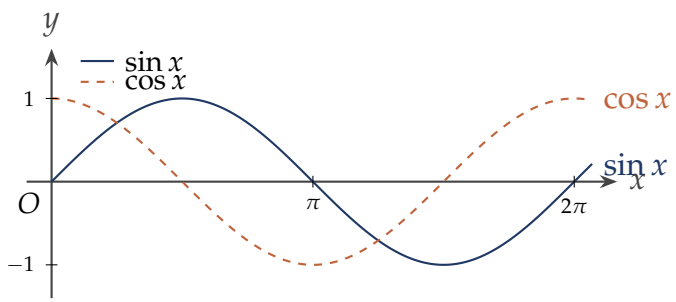

Figure 3. Plots of $\sin x$ and $\cos x$ over $[0, 2\pi]$ drawn with TikZ.

5.4 Sub-Figures

The `subcaption` package allows multiple figures side by side, each with its own label (see Figure 4).

(a) A filled circle. **(b)** A filled triangle.

Figure 4. Two geometric shapes as sub-figures: (a) circle and (b) triangle.

6 Code Listings

The `listings` package typesets source code with syntax colouring.

6.1 Python

```python
def newton(f, df, x0, tol=1e-9, max_iter=100):
    """
    Find a root of f using Newton-Raphson iteration.
    f   : function whose root is sought
    df  : derivative of f
    x0  : initial guess
    """
    x = x0
    for i in range(max_iter):
        fx = f(x)
        if abs(fx) < tol:
            return x, i         # (root approximation, iterations used)
        x -= fx / df(x)        # Newton update step
    raise RuntimeError("Did not converge after %d iterations." % max_iter)

# Example usage: find sqrt(2) as root of f(x) = x^2 - 2
import math
root, n = newton(lambda x: x**2 - 2,
                 lambda x: 2*x,
                 x0=1.0)
print(f"root = {root:.10f}  (exact: {math.sqrt(2):.10f})")
```

Listing 1. Newton's method in Python.

6.2 C

```c
#include <stdio.h>
#include <stdlib.h>

/* Merge two sorted sub-arrays of arr[]. */
void merge(int *arr, int l, int m, int r) {
    int n1 = m - l + 1, n2 = r - m;
    int *L = malloc(n1 * sizeof(int));
    int *R = malloc(n2 * sizeof(int));
```

```c
    for (int i = 0; i < n1; i++) L[i] = arr[l + i];
    for (int j = 0; j < n2; j++) R[j] = arr[m + 1 + j];
    int i = 0, j = 0, k = l;
    while (i < n1 && j < n2)
        arr[k++] = (L[i] <= R[j]) ? L[i++] : R[j++];
    while (i < n1) arr[k++] = L[i++];
    while (j < n2) arr[k++] = R[j++];
    free(L); free(R);
}

/* Recursive merge sort. */
void mergeSort(int *arr, int l, int r) {
    if (l < r) {
        int m = l + (r - l) / 2;
        mergeSort(arr, l, m);
        mergeSort(arr, m + 1, r);
        merge(arr, l, m, r);
    }
}
```

Listing 2. Merge sort in C.

6.3 LaTeX Code Snippet

```latex
\documentclass[12pt, a4paper]{article}
\usepackage{amsmath}

\begin{document}
  \title{My First Document}
  \author{A University Student}
  \date{\today}
  \maketitle

  Hello, \LaTeX! The integral
  \[
    \int_0^\infty e^{-x^2}\,\mathrm{d}x = \frac{\sqrt{\pi}}{2}
  \]
  is the Gaussian integral.
\end{document}
```

Listing 3. A minimal LaTeX document.

7 Cross-Referencing

LaTeX provides a powerful labelling system. Assign a label with `\label{key}` and reference it later with `\ref{key}` or `\eqref{key}` (for equations) or `\pageref{key}`.

- Section 3 (*Mathematics*) is on page 9.
- Euler's formula is Equation (1).
- The Cauchy–Schwarz table is Table 1.
- The flowchart is Figure 2.
- The Python listing is Listing 1.

The `hyperref` package (loaded in the preamble) automatically converts all these references into **clickable hyperlinks** in the PDF.

8 Multi-Column Layout

The `multicol` package allows flowing text across several columns, ideal for glossaries, indexes, or newspaper-style layouts.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

9 Special Environments

9.1 Information and Warning Boxes

Note

Always compile a LaTeX document **twice** (sometimes three times) so that cross-references and the table of contents are fully resolved. The first pass writes label data; the second reads it.

Warning

Never use the `eqnarray` environment — it has subtle spacing bugs. Use `align` (from amsmath) instead. This is a well-known pitfall for LaTeX beginners.

Tip

Use the `latexmk` command-line tool to automatically run LaTeX the correct number of times, including BibTeX/Biber passes. Install it once: `sudo apt install latexmk`.

9.2 Verbatim Text

The `verbatim` environment prints text *exactly* as typed, useful for showing command-line instructions:

```
$ pdflatex sample_document.tex
$ pdflatex sample_document.tex
$ bibtex   sample_document
$ pdflatex sample_document.tex
```

For inline `verbatim`, use: `\verb`CODE`||`.

9.3 `todonotes`: Draft Annotations

During drafting, you can leave inline reminders:

The `\todo{message}` command (from the `todonotes` package) adds a coloured note in the margin and compiles a list of all TODOs.

10 Bibliography and Citations

LaTeX integrates tightly with BibTeX for managing references.

10.1 Citation Styles

The `natbib` package provides flexible citation styles:

- Author-year: `\citep{knuth1984}` → (Knuth, 1984)
- Textual: `\citet{lamport1994}` → Lamport (1994)
- Numbered: with `style=numeric` in `natbib`

The bibliography entries are stored in a `.bib` file (see the `references.bib` file bundled with this sample). Include them at the end of the document with:

```
\bibliographystyle{plainnat}
\bibliography{references}
```

10.2 Sample `.bib` Entries

A `.bib` file uses the following format:

```bibtex
@book{knuth1984,
  author    = {Donald E. Knuth},
  title     = {The {\TeX}book},
  publisher = {Addison-Wesley},
  year      = {1984}
}

@book{lamport1994,
  author    = {Leslie Lamport},
  title     = {{\LaTeX}: A Document Preparation System},
  publisher = {Addison-Wesley},
  edition   = {2nd},
  year      = {1994}
}

@article{einstein1905,
  author  = {Albert Einstein},
  title   = {{Zur Elektrodynamik bewegter K{\"o}rper}},
  journal = {Annalen der Physik},
  volume  = {322},
```

```
  number   = {10},
  pages    = {891--921},
  year     = {1905}
}
```

Listing 4. Sample `.bib` entries.

11 Advanced Typography Tips

11.1 Dashes and Hyphens

LaTeX distinguishes three types of horizontal dashes:

Name	Source	Result
Hyphen	`-`	Self-reference
En dash	`-`	pages 10–20
Em dash	`--`	He paused—then spoke

11.2 Proper Quotation Marks

In LaTeX, *never* use the keyboard double-quote character directly. Instead:

- Single quotes: `'text'` → 'text'
- Double quotes: `''text''` → "text"

11.3 Non-Breaking Spaces and Fine Spacing

Use a tilde ~ for a non-breaking space (e.g. `Figure~\ref{...}`) so that the figure number never wraps to the next line.

Use `\,` for a thin space in formulae: $f\colon \mathbb{R} \to \mathbb{R}$ vs. $f : \mathbb{R} \to \mathbb{R}$.

11.4 Microtype Enhancements

The `microtype` package (loaded in the preamble) automatically applies character protrusion and font expansion, making paragraphs look more balanced with fewer hyphenation breaks — no user action required.

12 Organising Large Documents

`\input{file}`
: Inserts `file.tex` inline — errors are reported with line numbers relative to the *parent* file.

`\include{file}`
: Like `\input` but adds a `\clearpage` before/after and supports `\includeonly{}` for selective compilation (faster when working on large manuscripts).

Class `report` / `book`
: For multi-chapter documents use `report` (chapters, no recto/verso) or `book` (chapters, recto/verso layout, frontmatter/mainmatter/backmatter).

Version control
: Store your `.tex` and `.bib` files in Git. Add `*.aux`, `*.log`, `*.out`, `*.pdf` to `.gitignore`.

12.1 Quick Reference: What We Covered

Table 4. Summary of topics demonstrated in this document.

Topic	Packages / Commands Used
Text formatting	Core LaTeX, `xcolor`, `setspace`
Lists	`enumitem`
Mathematics	`amsmath`, `amssymb`, `amsthm`, `mathtools`
Tables	`booktabs`, `tabularx`, `multirow`, `xcolor` (rows)
Figures / graphics	`graphicx`, `tikz`, `subcaption`
Code listings	`listings`
Cross-referencing	`hyperref`, `\label`/`\ref`
Multi-column layout	`multicol`
Info/warning boxes	`tikz`
Bibliography	`natbib`, BibTeX
Typography	`microtype`, `fontenc`, `babel`
Large-doc organisation	`\input`/`\include`, document classes

A Useful LaTeX Resources

- Overleaf Documentation — the most beginner-friendly online reference.
- LaTeX Wikibook — comprehensive free textbook.
- CTAN (Comprehensive TeX Archive Network) — all LaTeX packages and their documentation.
- tex.stackexchange.com — the best Q&A community for LaTeX questions.
- Donald E. Knuth, *The TeXbook* (1984) — the definitive reference for the underlying TeX engine.
- Leslie Lamport, *LaTeX: A Document Preparation System* (2nd ed., 1994) — the original LaTeX manual.